

Design of PFC Boost Converter

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1 Introduction

The Power Factor Correction (PFC) boost converter is a type of DC-DC converter widely used in power supplies for its ability to provide a high power factor and low total harmonic distortion (THD). The PFC boost converter utilizes a boost topology to increase the output voltage. In this report, we describe the design and implementation of a PFC boost converter. The inductor used in this converter was calculated and wound by us in the lab.

2 PFC Boost Converter Specification

2.1 Specifications

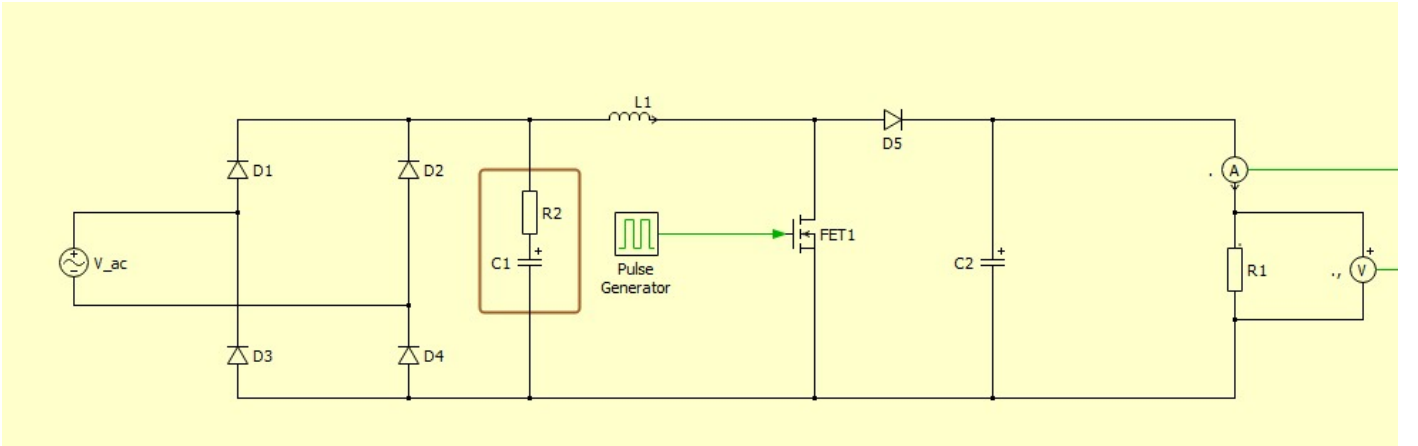


Figure 1: Boost PFC circuit in PLECS

Component/Ratings	Specification
V_{ac} (RMS)	35.35 V
C_1	500 μ F
L_1	881 μ H
F_s	10,000 Hz
C_2	400 μ F
R_1	150 Ω
Duty Cycle	0.4
$\frac{\Delta V_o}{\langle V_o \rangle}$	0.0346
$\langle V_o \rangle$	82 V

Table 1: Design Specifications

2.2 Inductor Design

The design parameters used for the inductor are given in the table below:

Parameters	Specification
I_{rms}	3.3 A
Inductance	881 μ H
J (current density for copper)	300 A/mm ²
K_w (window utilization factor)	0.4
B_m (maximum magnetic flux density)	0.2 T

Table 2: Inductor Design Specifications

To begin with, the first equation is given by:

$$A_p = A_c A_w = \frac{L I_p I_{rms}}{J K_w B_m} \quad (1)$$

Where:

- A_p is the area product,
- A_c is the area of cross-section,
- A_w is the window area,
- I_p is the peak current, which is equal to $\sqrt{2} \cdot I_{rms}$.

After substituting values from the table above, we get:

$$A_p = 4.793 \text{ cm}^4 \quad (2)$$

To calculate the area of the wire required (a_w), we use the following equation:

$$a_w = \frac{I_{rms}}{J} \quad (3)$$

Substituting the values from the table above, we get:

$$a_w = 0.011 \text{ cm}^2 \quad (4)$$

Considering these two values of A_p and a_w , we chose SWG-18 wire, and the core selected was EE 42/21/20. Diameter of SWG-18 wire (D) = 1.219 mm.

- A_p of the selected core = 60160 mm⁴,
- A_w of the selected core = 256 mm⁴,
- A_c of the selected core = 235 mm⁴,
- l_m (magnetic length) of the selected core = 98 mm.

Now, to calculate the number of turns, we use the following equation:

$$A_c = \frac{\text{Magnetic flux}}{\text{flux density (peak)}} = \frac{L I_p}{N B_m} \quad (5)$$

After substituting the values, we get the number of turns required:

$$N = \frac{0.881 \times 10^{-3} \times 3.3 \times \sqrt{2}}{0.2 \times 235 \times 10^{-6}} = 87.47 \text{ turns} \quad (6)$$

We know that:

$$\text{Magnetomotive Force (MMF)} = (\text{magnetic flux}) \cdot (\text{reluctance}) \quad (7)$$

This implies:

$$N I_p = B_m A_c \left(\frac{l_g}{\mu_0 A_c} + \frac{l_c}{\mu_0 \mu_r A_c} \right) \quad (8)$$

$$\mu_r \simeq 2000 \quad (9)$$

$$\text{thus, } \frac{l_g}{\mu_0 A_c} \gg \frac{l_c}{\mu_0 \mu_r A_c} \quad (10)$$

This leads to:

$$NI_p = B_m A_c \frac{l_g}{\mu_0 A_c} \quad (11)$$

After rearranging the above equation and canceling the terms, we get:

$$l_g = \frac{NI_p \mu_0}{B_m} \quad (12)$$

After substituting the values, we find:

$$l_g = \frac{4\pi \times 10^{-7} \times 87.47 \times 3.3 \times \sqrt{2}}{0.2} \quad (13)$$

Finally, we obtain:

$$l_g = 2.5648 \text{ mm} \quad (14)$$

Therefore, we conclude the calculation and modeling of the inductor. The number of turns required is 87.47 turns, and the length of the gap is 2.5648 mm.

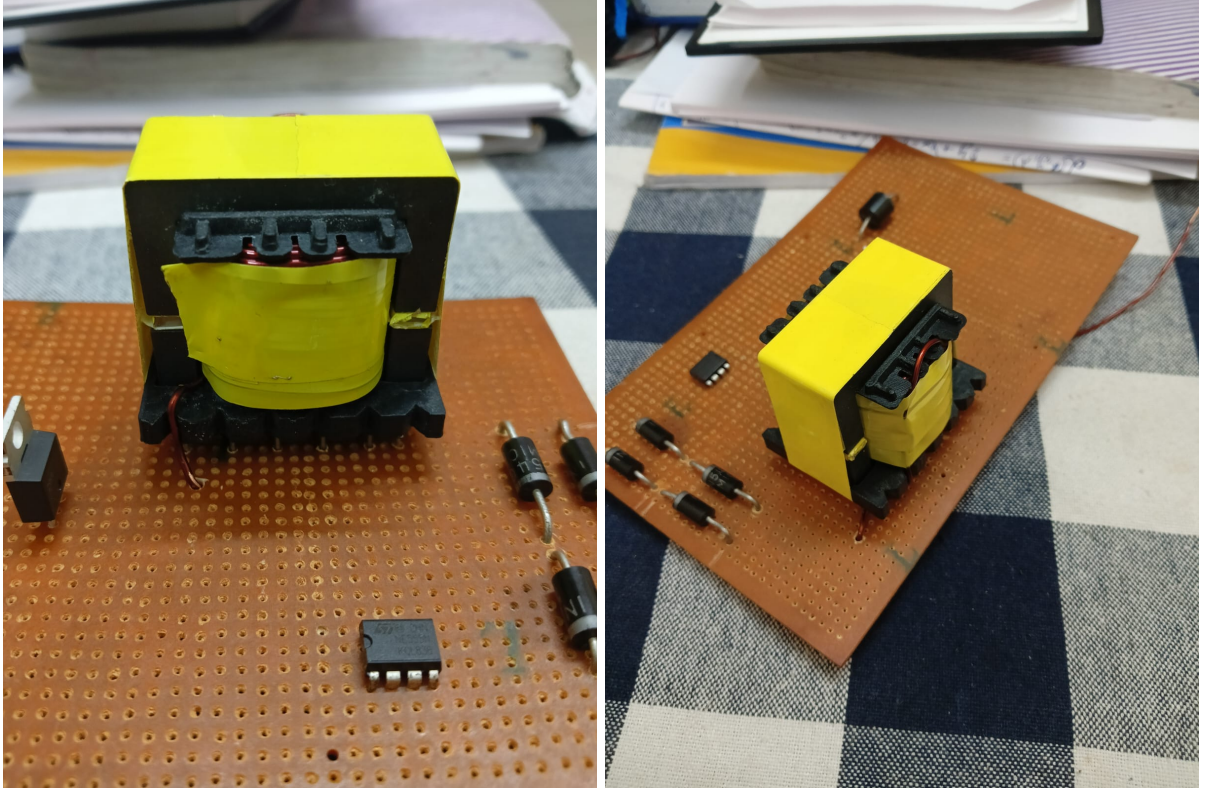


Figure 2: The final modeled inductor

2.3 Simulation Results

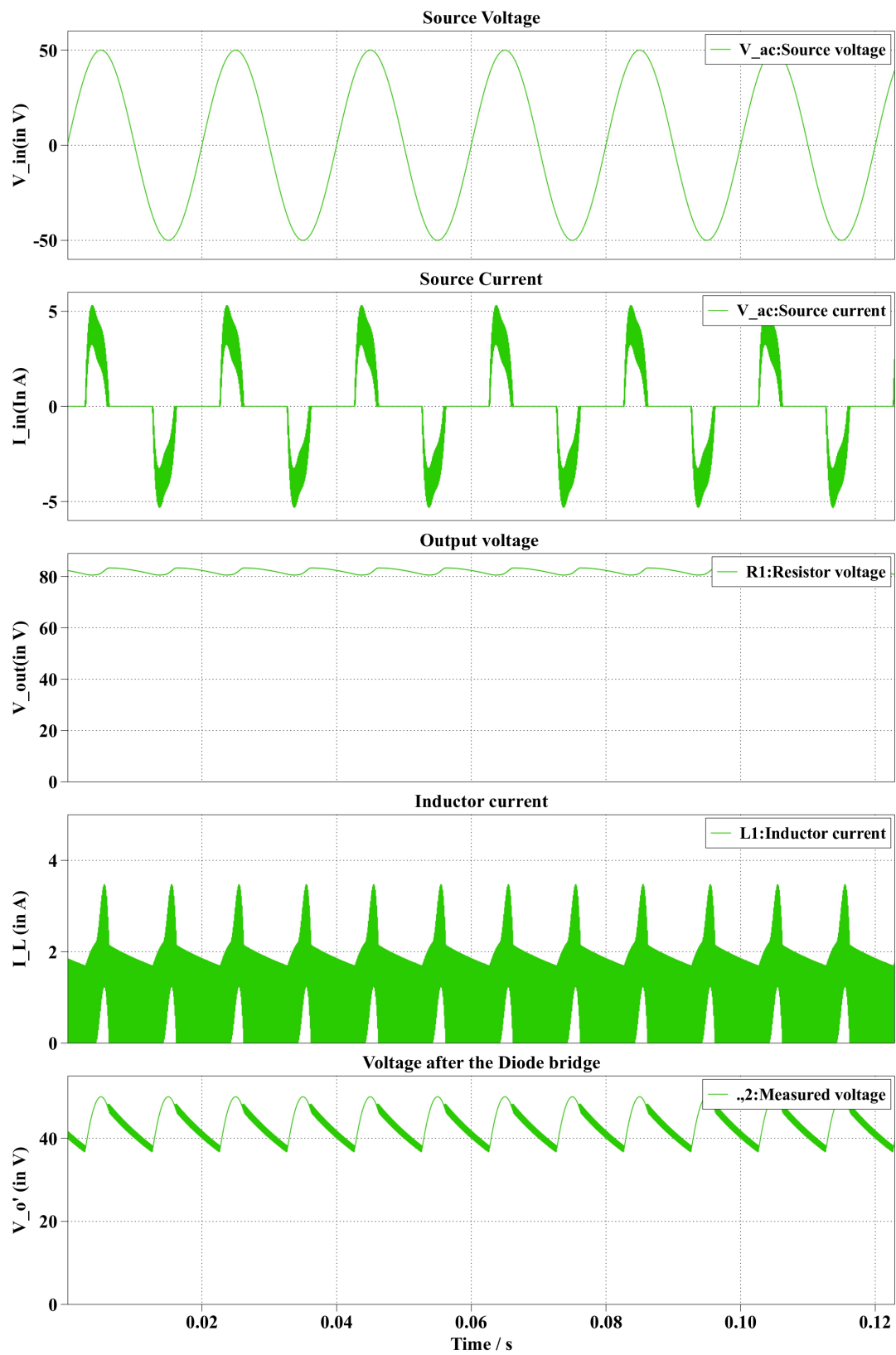


Figure 3: Simulation results obtained from the above circuit

Parameter	Specifications
I_{in} (peak)	5.32 A
$\langle V_{\text{out}} \rangle$	82.065 V
I_L (peak)	3.477 A
$\langle I_L \rangle$	1 A
$\langle V_o' \rangle$ (Voltage after bridge rectifier)	43.7 V

Table 3: Results Table

2.4 Results and Discussion

Some of the conclusions that we can draw from the above simulation are:

- For constant pulses with a duty ratio of 0.4 and a switching frequency of 10 kHz, the designed PFC circuit operates in Discontinuous Conduction Mode (DCM).
- The percentage ripple in the output voltage is 3.5%.
- The output voltage measured is 82 V.
- The power factor of the circuit was found to be 0.713.
- The on-state resistance of the post-diode bridge capacitor (R_2) is assumed to be 1Ω because the voltage cannot be directly applied to the capacitor; hence, the resistance is placed to simulate the circuit.
- The efficiency of the converter is calculated to be 95.43%.